

These notes are from a chat with Alan Flett. The goal was to see how can we make use of SHEO in context of Semaphore. These notes shine bright light on the nature of the impedance mismatch.

## Summary

**Alan Flett:** professional services consultant at Smart Logic. Before, did a lot of ontology stuff, enterprise modeling, description logics etc. both in academia and commercial.

Alan Flett is a consultant, rather than the one who designed the guts of the metamodel. I asked if the metamodel was well documented either in say a UML diagram or a 10ish page report. No such luck. Fortunately, Alan has solid understanding of description logics, so we were communicating at high bandwidth. He confirmed that they are firmly in 'term land' and 'document land' vs. 'concept'-land which is where SHEO (and OWL for the most part) lives. But that there are things in the Semaphore ontology metamodel that are very different from most term knowledge structuring systems, that make it a strange hybrid. For example, domain and range for relationships, which have direct analogs in OWL properties, but whose semantics is very different.

In term/document/search land, the prime focus is if the user types in a given term, and there are other terms with related meaning, how much *evidence* is there that if the user enters one as a search term, and the document has the other term, that they will be interested in that document even though their search term is not explicitly mentioned. The degenerate case of this is semantically synonymous terms like auto and car, where the evidence should be 100%.

The user can create other relationships like *partOf*, *instanceOf*, etc that seem just like OWL properties. The difference to always keep in mind is that a property in SHEO specifies a particular way that two things may be related to each other in the real world of Sentara Healthcare. In Semaphore, a property allows one can say how a term can be used as evidence for another term. It also specifies certain behaviors in the UI. There are three main kinds of relationship in Semaphore, each having a different impact on the classification and navigation, and each (presumably) reflecting some real world relationship between the terms and the concepts that the terms are implicitly referring to. The strength of the connection of the real world terms to each other will increase the evidence that one term is of interest to the searcher if they type in the other term.

- a. **hierarchical:** between any two *preferred terms*, one cannot hierarchically relate non-preferred terms; there is one hierarchical relationship that is built in: BT/NT, and it is generally the case that no user-created hierarchical relationships are created. Hierarchical relationships show up in a search trail 'bread crumbs'. There are default rules that specify that if one is search for a parent or child, then the child or parent will show up somehow.
- b. **associative:** between any two *preferred terms*, cannot associate non-preferred terms. These are used to specify evidence for the relevance of one term in a document given

another term is present in the document. The RT relationship is a built-in associative relationship, (same as the SKOS:relatedTo; it comes with default behavior in terms of content indexing. If you want to specify other relationships that have different behavior in terms of evidence in content indexing, you can create user-defined associative relationships. E.g. you might want to say that `partOf` vs. `subTopicOf` have different degrees of strength in terms of evidence.

- c. **equivalence:** this is really a misnomer, as it does not mean semantic equivalence at all. It is a bit like query expansion, if one given term is searched for, then the other terms related by an equivalence relationship to the search term are added to the search term as if they had been typed in, in the first place. So this is maybe why they call the relationship equivalence, you could have typed any of the equivalence terms in and gotten the same search result?

Alan says that the limitation to preferred terms is probably a bad idea.

An important choice is whether to model something as a preferred vs. non-preferred term. This apparently has nothing to do with what anything means, it has only to do with whether the term shows up enough in the documents *and* is important enough that users will likely use it as a search term. In SHEO there is a class: `sheo:Hospital`, and we could create an instance called `shei:NorfolkGeneral`. A particular hospital is a narrower concept than the concept of Hospital in general, so if we want to capture this in the Semaphor ontology, we create a term called Hospital and a term called Norfolk General. Let's say we know that Hospital is a preferred term, and we people search for documents with one term, they might be interested in documents that mention the other term. Here are two alternative ways to 'transform' the sheo terms and see how they end up in a Semaphor ontology. First: if Norfolk General is important and will likely be used as a search term, then you will want to make Norfolk General a preferred term that is NT Hospital (using the builtin hierarchical relationship BT/NT). Second, if Norfolk General is unlikely to be used as a search term, but we want to return documents that mention it if one is searching for documents about hospitals, and you want it to show up in a list of related things for navigation purposes then we make Norfolk General a non-preferred term and use an *equivalence* relationship to associate Hospital with Norfolk General. Here we might create an equivalence relationship called *isInstanceOf* which connects the non preferred term, Norfolk General to the preferred term Hospital. Any two things related by this equivalence relationship will be treated as partial evidence. One can weight different relationships differently in accordance to how the real-world meaning of the relationship impacts the non-preferred term being evidence for the preferred term. From Semaphor's perspective, that is all it means. In OWL there is potentially a lot more meaning that can be given to a property (transitivity etc).

Note that in the first case, we have no direct hint in Semaphor that Norfolk General is an instance of Hospital, only that it is a NT of Hospital. We can do that by adding an associative relationship called *isInstanceOf* and relating them with that as well as the NT hierarchical relationship. (I don't know if you can use the same name for both an equivalence relationship and an associative relationship, let's say we can for this discussion). So now, you have the same two things Hospital and Norfolk General related by BT/NT which is a hierarchical relationship, and by *isInstanceOf* which is an associative relationship.

Note that the two *isInstanceOf* relationships (associative vs. equivalence) semantically mean exactly the same thing from real world perspective, but they are treated very differently by Semaphor (i.e., they have different *operational semantics* from a Semaphor perspective. You will get different context indexing and search results and UI behavior. Exactly what the difference is, is the \$million question that is not documented well.

There are other real world relationships that semantically ought to map to BT/NT such as *partOf* and *subTopicOf* but whose real world semantics is very different. In OWL, you would want to make them subproperties of BT/NT, but that is not supported in Semaphor.

In the case of *partOf*, again if the narrower (part) term is likely to be used as a search term, then it will be a preferred term and related to the broader terms using BT/NT. If you want different behavior you can also create an associative relationship called *partOf* and specify rules for evidence gathering that will compliment the rules used for the BT/NT hierarchy. NB: this is speculative as to whether it makes sense to do this, it would have to be tried out and/or confirmed with say, Alan.

Again, if the part term is *not* likely to be used as a search term, you can create an equivalence relationship called *partOf* and use it to relate the part to the 'whole'. This is the same as for the *isInstanceOf* case described above.

Now you have two different relationships that logically belong in the BT/NT hierarchy: *instanceOf* and *partOf*. But there is no subproperty in Semaphor, so you have to create the link twice in each case. If evidence for one kind of relationship is different from another, then you can *also* create an associative property e.g. called *isInstanceOf* and use it to relate the two preferred terms.

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Preferred terms belong to classes. This is a bit weird, still not sure what class means, Alan said more like subclass in OWL than subclass in OO, but as he talked about it, it seemed more like the latter. Classes are associated with top level preferred terms and each new class has a template for defining rules to use for evidence.